Operations & Supply Chain Management

Final Assignment

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Question 1: Facility Location Analysis

Centre of Gravity Method

The center of gravity method provides a quantitative approach to facility location decisions, specifically when the primary objective involves minimizing transportation costs associated with serving multiple areas or facilities (International Six Sigma Inc., 2011). This method computes geographic coordinates for a potential single new facility that will minimize costs by considering demand volumes and geographical coordinates.

Calculation for Centre of Gravity:

Given data for existing agents:

- A-Seattle: (-122.3, 47.6) Weight: 60
- B-Atlanta: (-84.4, 33.7) Weight: 70
- C-Philadelphia: (-75.1, 40) Weight: 80
- D-Shanghai: (121.5, 31.2) Weight: 150
- E-Berlin: (13.4, 52.5) Weight: 170

The centre of gravity coordinates are calculated using:

- X-coordinate = $\Sigma (Wi \times Xi) / \Sigma Wi$
- Y-coordinate = $\Sigma (Wi \times Yi) / \Sigma Wi$

X-coordinate calculation: X = [60(-122.3) + 70(-84.4) + 80(-75.1) + 150(121.5) + 170(13.4)] / (60+70+80+150+170) X = [-7,338 - 5,908 - 6,008 + 18,225 + 2,278] / 530 X = 1,249 / 530 = **2.36**

Y-coordinate calculation: Y = [60(47.6) + 70(33.7) + 80(40) + 150(31.2) + 170(52.5)] / 530 Y = [2,856 + 2,359 + 3,200 + 4,680 + 8,925] / 530 Y = 22,020 / 530 = **41.55**

Optimal facility location: (2.36, 41.55)

Load-Distance Method Analysis

The load-distance method evaluates potential sites using Euclidean distance calculations. For each potential site, distances to all existing agents are calculated and multiplied by their respective weights.

Potential sites:

- 1-Athens: (23.7, 37.9)
- 2-Amsterdam: (5, 52.2)
- 3-Shanghai: (121.5, 31.2)

Distance calculations using Euclidean formula: $\sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$

For Athens (23.7, 37.9):

- Distance to Seattle: $\sqrt{[(-122.3-23.7)^2 + (47.6-37.9)^2]} = 146.3$
- Distance to Atlanta: $\sqrt{[(-84.4-23.7)^2 + (33.7-37.9)^2]} = 108.3$
- Distance to Philadelphia: $\sqrt{[(-75.1-23.7)^2 + (40-37.9)^2]} = 98.8$
- Distance to Shanghai: $\sqrt{[(121.5-23.7)^2 + (31.2-37.9)^2]} = 98.0$
- Distance to Berlin: $\sqrt{[(13.4-23.7)^2 + (52.5-37.9)^2]} = 17.6$

Total load-distance for Athens: 60(146.3) + 70(108.3) + 80(98.8) + 150(98.0) + 170(17.6) = **44,368**

For Amsterdam (5, 52.2):

- Distance to Seattle: $\sqrt{[(-122.3-5)^2 + (47.6-52.2)^2]} = 127.4$
- Distance to Atlanta: $\sqrt{[(-84.4-5)^2 + (33.7-52.2)^2]} = 91.3$
- Distance to Philadelphia: $\sqrt{[(-75.1-5)^2 + (40-52.2)^2]} = 81.0$
- Distance to Shanghai: $\sqrt{[(121.5-5)^2 + (31.2-52.2)^2]} = 118.4$
- Distance to Berlin: $\sqrt{[(13.4-5)^2 + (52.5-52.2)^2]} = 8.4$

Total load-distance for Amsterdam: 60(127.4) + 70(91.3) + 80(81.0) + 150(118.4) + 170(8.4) = **36,638**

For Shanghai (121.5, 31.2):

- Distance to Seattle: $\sqrt{[(-122.3-121.5)^2 + (47.6-31.2)^2]} = 244.4$
- Distance to Atlanta: $\sqrt{[(-84.4-121.5)^2 + (33.7-31.2)^2]} = 206.0$
- Distance to Philadelphia: $\sqrt{[(-75.1-121.5)^2 + (40-31.2)^2]} = 196.8$
- Distance to Shanghai: $\sqrt{[(121.5-121.5)^2 + (31.2-31.2)^2]} = 0$
- Distance to Berlin: $\sqrt{[(13.4-121.5)^2 + (52.5-31.2)^2]} = 110.1$

Total load-distance for Shanghai: 60(244.4) + 70(206.0) + 80(196.8) + 150(0) + 170(110.1) = **48,981**

Best potential site: Amsterdam with the lowest total load-distance of 36,638.

Question 2: Aggregate Planning and Productivity Analysis

a) Aggregate Planning in Service Industries vs Manufacturing

Aggregate planning in service industries presents unique challenges compared to manufacturing environments. Service businesses face distinct operational complexities that manufacturing companies typically do not encounter (Inman, 2006).

Advantages in service industries:

- Enhanced customer satisfaction through capacity optimization
- Improved resource utilization during peak and off-peak periods
- Better workforce scheduling flexibility
- Reduced operational costs through strategic planning
- Enhanced ability to manage seasonal demand variations

Challenges in service industries:

- Services cannot be stored or inventoried, requiring real-time capacity matching
- Higher demand variability and uncertainty

- Customer involvement in service delivery processes
- Difficulty in standardizing service outputs
- Quality control complexities due to human factors
- Location-dependent demand patterns

Manufacturing aggregate planning typically involves tangible products that can be stored, while service planning must account for the intangible nature of services and immediate consumption requirements (The Intact One, 2019).

b) Productivity Measures and Calculations

Single-factor productivity measures output relative to one input factor, such as labor productivity or material productivity.

Multi-factor productivity measures output relative to multiple input factors combined, such as labor and materials.

Total-factor productivity measures output relative to all input factors including labor, materials, capital, and energy.

Given data:

- Production: 180 pieces at Rs 500 each = Rs 90,000
- Labor cost: Rs 4,000
- Material cost: Rs 20,000
- Overhead: Rs 12,000
- Total input cost: Rs 36,000

Multi-factor productivity calculation: Multi-factor productivity = Total output value / (Labor cost + Material cost + Overhead) Multi-factor productivity = Rs 90,000 / (Rs 4,000 + Rs 20,000 + Rs 12,000) Multi-factor productivity = Rs 90,000 / Rs 36,000 = **2.5**

The multi-factor productivity equals 2.5, indicating that for every rupee invested in combined inputs, Rs 2.5 worth of output value is generated.

Question 3: Assembly Line Balancing

Real Fruit Snack requires producing 6,000 fruit strips in a 40-hour week, establishing a production rate requirement of 150 strips per hour or one strip every 0.4 minutes (24 seconds).

Given work elements:

- A: Press out sheet (0.1 min, no precedence)
- B: Cut into strips (0.2 min, follows A)
- C: Outline shapes (0.4 min, follows A)
- D: Roll and package (0.3 min, follows B and C)

Total task time: 0.1 + 0.2 + 0.4 + 0.3 = 1.0 minute

Theoretical minimum workstations: 1.0 min / 0.4 min = 2.5 ≈ 3 workstations

Assembly line design:

Workstation 1: Task A (0.1 min) + Task B (0.2 min) = 0.3 min **Workstation 2:** Task C (0.4 min) **Workstation 3:** Task D (0.3 min)

This design achieves the production quota with three workstations while respecting all precedence constraints. The cycle time of 0.4 minutes per workstation ensures meeting the demand requirement of 6,000 strips per 40-hour week.

Question 4: Forecasting Methods

Historical data:

- Period 1: 60 complaints
- Period 2: 65 complaints
- Period 3: 55 complaints
- Period 4: 58 complaints
- Period 5: 64 complaints

1. Naive Approach

The naive approach uses the most recent period's value as the forecast. **Period 6 forecast: 64 complaints**

2. Three-period Moving Average

Moving average = (Period 3 + Period 4 + Period 5) / 3 **Period 6 forecast:** (55 + 58 + 64) / 3 = **59 complaints**

3. Weighted Average

Weighted average = 0.50(64) + 0.30(58) + 0.20(55) **Period 6 forecast:** 32 + 17.4 + 11 = **60.4 complaints**

4. Exponential Smoothing ($\alpha = 0.40$)

Formula: $F_{t+1} = \alpha A_t + (1-\alpha) F_t$

Working backwards from Period 1:

- F2 = 0.40(60) + 0.60(60) = 60
- F3 = 0.40(65) + 0.60(60) = 62
- F4 = 0.40(55) + 0.60(62) = 59.2
- F5 = 0.40(58) + 0.60(59.2) = 58.72
- F6 = 0.40(64) + 0.60(58.72) = 60.83

Period 6 forecast: 60.83 complaints

Question 5: Job Sequencing Analysis

Job data:

Job data (arrival = time 0 for all jobs)

- Job A: 2 days processing, due day 7
- Job B: 8 days processing, due day 18
- Job C: 4 days processing, due day 4

- Job D: 10 days processing, due day 17
- Job E: 5 days processing, due day 15
- Job F: 12 days processing, due day 18

Makespan (total time to finish all jobs) = **41 days** for every schedule (sum of all processing times).

Important note: Regardless of the sequencing rule applied, the **makespan (total time to complete all jobs) remains constant at 41 days** for every scheduling method. This is because makespan equals the sum of all processing times (2 + 8 + 4 + 10 + 5 + 12 = 41 days), which is independent of job sequence. The scheduling rules only affect the order in which jobs are completed, thereby influencing individual job completion times, flow times, and tardiness measures, but not the overall project duration.

1. FCFS (First Come, First Served)

Sequence: A - B - C - D - E - F

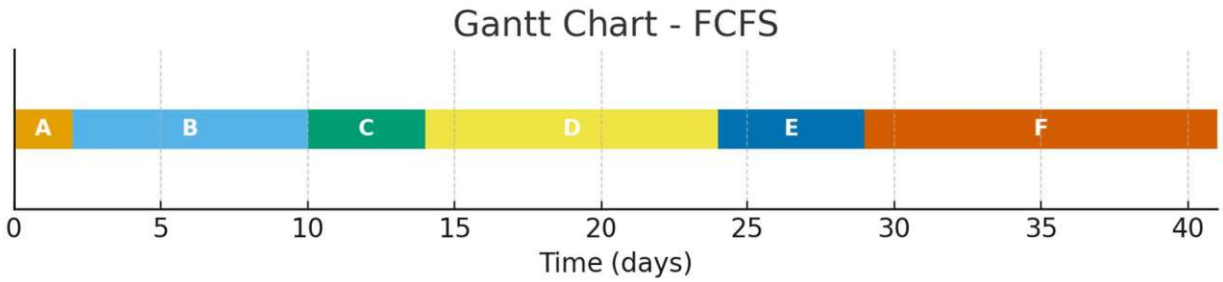
Per-job table (Start, Completion, Flow time = completion, Due, Tardiness = max (0, completion − due)):

Job	PT	Start	Completion	Flow	Due	Tardiness
A	2	0	2	2	7	0
B	8	2	10	10	18	0
C	4	10	14	14	4	10
D	10	14	24	24	17	7
E	5	24	29	29	15	14
F	12	29	41	41	18	23

Totals / averages:

- Sum of flow times = 120
- Average flow time = 120 / 6 = **20.00 days**
- Sum of tardiness = 54

- Average tardiness = 54 / 6 = **9.00 days**
- Average number of jobs at work center = Sum flow / makespan = 120 / 41 = **2.93 jobs**



2. SPT (Shortest Processing Time)

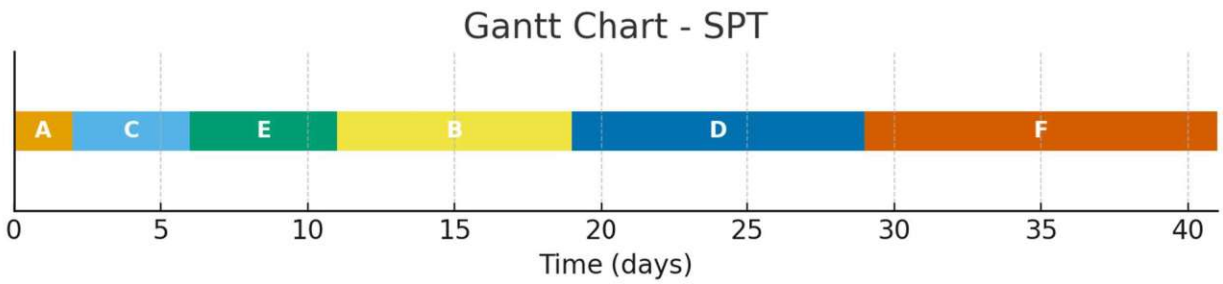
Sequence: A - C - E - B - D - F

Per-job table:

Job	PT	Start	Completion	Flow	Due	Tardiness
A	2	0	2	2	7	0
C	4	2	6	6	4	2
E	5	6	11	11	15	0
B	8	11	19	19	18	1
D	10	19	29	29	17	12
F	12	29	41	41	18	23

Totals / averages:

- Sum of flow times = 108
- Average flow time = 108 / 6 = **18.00 days** (lowest)
- Sum of tardiness = 38
- Average tardiness = 38 / 6 = **6.33 days**
- Average number of jobs = 108 / 41 = **2.63 jobs**



3. EDD (Earliest Due Date)

Sequence: C - A - E - D - B - F

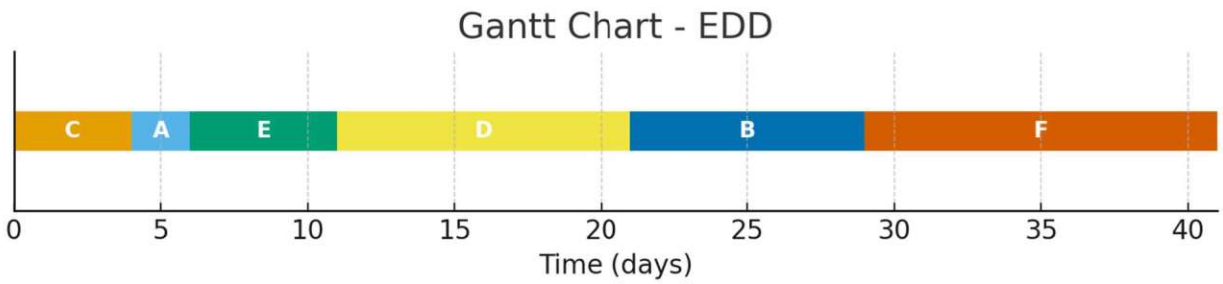
(If tie-breaking were needed for B and F (both due day 18), arrival order is used — B before F.)

Per-job table:

Job	PT	Start	Completion	Flow	Due	Tardiness
C	4	0	4	4	4	0
A	2	4	6	6	7	0
E	5	6	11	11	15	0
D	10	11	21	21	17	4
B	8	21	29	29	18	11
F	12	29	41	41	18	23

Totals / averages:

- Sum of flow times = 112
- Average flow time = 112 / 6 = **18.67 days**
- Sum of tardiness = 38
- Average tardiness = 38 / 6 = **6.33 days** (ties with SPT)
- Average number of jobs = 112 / 41 = **2.73 jobs**



4. CR (Critical Ratio) – Static Interpretation (compute critical ratio at time 0 and sort by that)

Critical ratio at time 0 = (Due − t) / PT with t = 0:

- A: 7 / 2 = 3.50
- B: 18 / 8 = 2.25
- C: 4 / 4 = 1.00
- D: 17 /10 = 1.70
- E: 15 / 5 = 3.00
- F: 18 /12 = 1.50

Sequence (ascending CR = most urgent first): C - F - D - B - E - A

Per-job table:

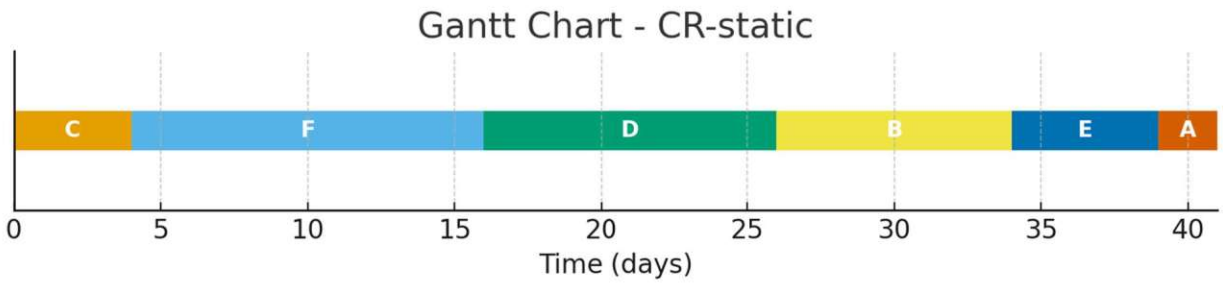
Job	PT	Start	Completion	Flow	Due	Tardiness
C	4	0	4	4	4	0
F	12	4	16	16	18	0
D	10	16	26	26	17	9
B	8	26	34	34	18	16
E	5	34	39	39	15	24
A	2	39	41	41	7	34

Totals / averages:

- Sum of flow times = 160

- Average flow time = 160 / 6 = **26.67 days**
- Sum of tardiness = 83
- Average tardiness = 83 / 6 = **13.83 days**
- Average number of jobs = 160 / 41 = **3.90 jobs**

| *Note: this is the "static CR" interpretation (CRs computed once at t = 0 and sorted).*



5. CR — dynamic interpretation (recompute CR at each decision point)

At each decision time t, compute CR = (Due − t) / PT for all remaining jobs; pick the job with smallest CR (most urgent). With this rule the selected sequence is:

Sequence (dynamic CR): C - F - A - E - B - D

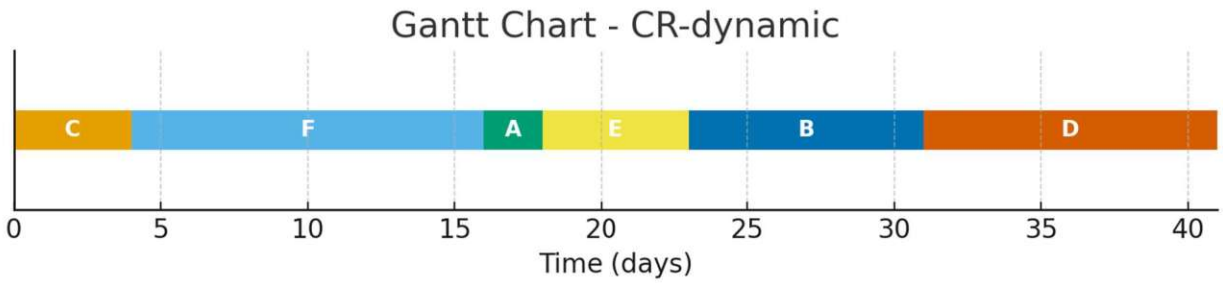
Per-job table:

Job	PT	Start	Completion	Flow	Due	Tardiness
C	4	0	4	4	4	0
F	12	4	16	16	18	0
A	2	16	18	18	7	11
E	5	18	23	23	15	8
B	8	23	31	31	18	13
D	10	31	41	41	17	24

Totals / averages:

- Sum of flow times = 133
- Average flow time = 133 / 6 = **22.17 days**

- Sum of tardiness = 56
- Average tardiness = 56 / 6 = **9.33 days**
- Average number of jobs = 133 / 41 = **3.24 jobs**



Optimal rule:

1. SPT provides the best performance with lowest average flow time (18 days) and average tardiness (6.33 days), while maintaining reasonable work center utilization (2.63 jobs).
2. **SPT and EDD tie on average tardiness** (both total tardiness = 38 → average 6.33 days). SPT does **not** uniquely minimize average tardiness.
3. The "CR" rule can be interpreted either **statically** (compute CRs once at t = 0) or **dynamically** (recompute after each completion). Static and dynamic interpretations produce different sequences and different performance measures.

Practical applications and method selection

The choice of scheduling rules depends on organizational priorities and operational context. **FCFS (First Come, First Served)** is preferred when maintaining fairness and customer service equity is paramount, legal or contractual obligations require order-based processing, minimal computational resources are available for scheduling optimization, or customer relationships are prioritized over operational efficiency. **SPT (Shortest Processing Time)** is optimal when minimizing work-in-process inventory and associated carrying costs, maximizing throughput and resource utilization, reducing average customer waiting times in service environments, or operating under capacity constraints where quick turnaround is critical. **EDD (Earliest Due Date)** is most suitable when meeting promised delivery dates is the primary competitive advantage, penalty costs for late deliveries are significant, customer satisfaction depends heavily on on-time performance or operating in environments with firm deadline commitments. **Critical Ratio (CR)** methods are appropriate when operating in dynamic

environments where priorities frequently change, processing complex multi-stage operations requiring adaptive scheduling, balancing urgency with processing efficiency considerations, or managing mixed-model production with varying customer priorities (Naji Nasrabadi Yazd et al., 2023). In practice, many organizations implement **hybrid approaches**, combining multiple rules based on job characteristics, customer priorities, or operational phases. For instance, using EDD for high-priority customers while applying SPT for standard orders, or switching between methods based on current workload and capacity utilization levels.

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Question 1: Facility Location Analysis

Centre of Gravity Method & Load-Distance Analysis

Centre of Gravity Method

$$X = \Sigma(W_i \times X_i) / \Sigma W_i$$

$$Y = \Sigma(W_i \times Y_i) / \Sigma W_i$$

Optimal Location

(2.36, 41.55)

Load-Distance Results

Site	Total Load-Distance
Athens	44,368
Amsterdam	36,638
Shanghai	48,981

Recommendation: Amsterdam is the optimal location among potential sites with the lowest total load-distance of 36,638, offering best accessibility to all agents.

Question 2: Aggregate Planning & Productivity

Service vs Manufacturing Analysis

Service Industry Advantages

- Enhanced customer satisfaction
- Resource optimization
- Workforce flexibility
- Cost reduction

Service Industry Challenges

- Cannot inventory services
- Higher demand variability
- Customer involvement complexity
- Quality control difficulties

Multi-Factor Productivity Calculation

```
MFP = Output Value / (Labor + Materials + Overhead)
MFP = Rs 90,000 / Rs 36,000
```

2.5

For every rupee invested, Rs 2.5 worth of output is generated

Question 3: Assembly Line Balancing

Real Fruit Snack Production Line Design

Production Requirements

6,000

strips per 40-hour week

0.4 min

cycle time required

Task Analysis

Optimal Assembly Line Design

Station 1	Station 2	Station 3
Tasks A + B	Task C	Task D
0.3 minutes	0.4 minutes	0.3 minutes

Result: 3 workstations meet production quota while respecting precedence constraints

Question 4: Forecasting Methods

Period 6 Complaint Forecasts

Naive Approach

64

complaints

3-Period Moving Average

59

complaints

Weighted Average

60.4

complaints

Exponential Smoothing

60.8

complaints

Historical Data: Period 1: 60, Period 2: 65, Period 3: 55, Period 4: 58, Period 5: 64

Question 5: Job Sequencing Analysis

Scheduling Rule Comparison

Scheduling Rule	Sequence	Avg Flow Time	Avg Tardiness	Avg Jobs in System
FCFS	A-B-C-D-E-F	20.0 days	9.0 days	2.93 jobs
SPT	**A-C-E-B-D-F**	**18.0 days**	**6.33 days**	**2.63 jobs**
EDD	C-A-E-D-B-F	18.67 days	6.33 days	2.73 jobs
CR	C-F-D-B-E-A	26.67 days	13.83 days	3.90 jobs

Optimal Scheduling Rule

SPT

Shortest Processing Time provides best overall performance with lowest flow time and tardiness

Key Insight: SPT minimizes work-in-process inventory and reduces customer waiting times, making it ideal for this job mix.

Executive Summary

Key Findings & Recommendations

🎯 Facility Location

Amsterdam optimal site (Load-Distance: 36,638)
Centre of Gravity: (2.36, 41.55)

📊 Productivity

Multi-factor Productivity: 2.5
Services face unique planning challenges vs manufacturing

🏭 Assembly Line

3 Workstations meet 6,000 strips/week target
Balanced design respects all precedence constraints

📈 Forecasting

Period 6 forecasts range: **59-64 complaints**
Multiple methods provide forecast validation

⏰ Job Scheduling

SPT Rule optimal: 18 days flow time
6.33 days average tardiness, 2.63 jobs in system

💡 Strategic Insights

Quantitative methods enable data-driven decisions
Multiple evaluation criteria ensure robust solutions